NGTA: An Auditable Neurosymbolic Interface for Clinical Transformers with Incomplete Inputs

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Abstract

Dynamic evidential routing is an inference-time interface for uncertainty-conditioned symbolic intervention. Prototype rules revise a heuristic confidence signal used by a transformer readout. We evaluate 20 hospital-held-out WiDS fits across two imputers, APACHE inclusion/exclusion, and five training seeds, plus 15 clinical-only, genomic-only, and fused TCGA fits using checksum-pinned public mutation files. Every fit uses matched 50-pass dropout inference with three independent repeats and independently replayable intervention exports. The acquired TCGA test cases include 39 recorded BRAF-positive rule events, while absent variants remain unknown without callable assay evidence. Hierarchical comparisons do not meet the prespecified symbolic Brier benefit of 10^{-4}; APACHE removal substantially reduces discrimination. Frozen masking and ensemble diagnostics test implementation behavior on development cohorts. External compatibility, clinical rule validity, and reviewer benefit remain unestablished.

1 Introduction

Clinical AI systems, including emerging generative and agentic systems, need an inspectable way to represent weak evidence and expert intervention. A sparse ICU record may contain a few abnormal measurements, many unmeasured variables, and an apparently precise risk score. Missingness, irregular measurement, and cohort shift are common in hospital data, yet standard neural predictors usually collapse the available evidence into one probability [Li et al., 2019b, Ovadia et al., 2019, Ghassemi et al., 2020]. Any downstream assistant that consumes that score inherits its hidden uncertainty, making an overconfident error difficult to defer, audit, or correct.

The challenge is that current model components expose different pieces of the problem but do not connect them cleanly. Transformers are flexible function approximators for clinical prediction [Rasmy et al., 2021, Azizi et al., 2021, Li et al., 2019a], but attention weights are not themselves evidential confidences. Uncertainty estimators can summarize prediction instability, but they usually do not provide a symbolic intervention point. Rule-based explanations can be readable, but post-hoc rules often describe a prediction without changing the computation. What is missing is an inference-time interface in which neural uncertainty, symbolic evidence, and attention control share an auditable representation.

We implement NGTA as an inference-time interface. Attention means and variances initialize feature-level truth values (f, c), borrowing the NAL frequency/confidence representation [Wang, 1995, 2006, Wang and Awan, 2011]. Explicit clinical rules then revise feature-level truths in the same NAL truth-value space, and the revised confidence values control a normalized attention gate. The signal is a heuristic interface variable; its relationship with errors and evidential support must be measured before it can serve as a trust signal.

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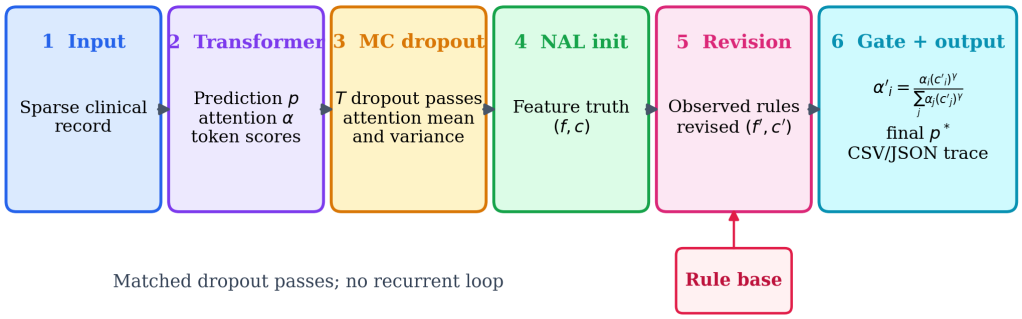


Figure 1: Inference-time dynamic evidential routing. A trained transformer produces a prediction and feature attention; MC dropout estimates feature attention means and variances; these initialize heuristic NAL truth values; triggered clinical rules revise those values; revised confidence gates attention; and the system exports a case-level trace showing where symbolic evidence changed the computation.

We evaluate the interface on two complementary clinical benchmarks. Acquired TCGA-THCA mutation files supply recorded held-out positives, but absence of variants is still unknown without verified callable assays. WiDS exercises scale on held-out hospitals with feature-specific missingness; selected-feature missingness is 10.36% after invalid APACHE probabilities are masked. The contribution is an auditable inference-time interface: neural uncertainty and explicit rules share a truth-value representation, revised confidence can change attention, and the arithmetic is exported. The experiments do not establish a clinical performance improvement, and they do not isolate symbolic revision as the source of any calibration difference.

Figure 1 summarizes the inference-time data flow from sparse clinical inputs through uncertainty estimation, NAL-style truth initialization, rule-triggered revision, confidence-gated attention, and trace export.

2 Related Work

Neurosymbolic integration. Neurosymbolic AI combines neural pattern recognition with symbolic reasoning, knowledge injection, or interpretable rule systems [d'Avila Garcez and Lamb, 2023, d'Avila Garcez et al., 2019, Bhuyan et al., 2024, Colelough and Regli, 2025]. Prior surveys distinguish loose coupling, tight coupling, and unified architectures [Hilario, 1997, Nawaz et al., 2025]. Dynamic evidential routing is a tight inference-time interface: the neural model supplies probabilities, uncertainty estimates, and attention weights, while the symbolic layer revises evidential confidence before attention is updated. Its contribution is the interface contract between these components. Neural uncertainty is converted into an explicit truth-value representation, symbolic rules revise that representation, and the revised confidence is fed back into attention with a trace of the intervention.

Clinical transformers and uncertainty. Clinical transformers have been used for structured EHRs, clinical text, and medical prediction tasks [Rasmy et al., 2021, Li et al., 2019a, Dai et al., 2025]. Attention can help interpretability, but medical attention mechanisms rarely represent uncertainty as a first-class control signal [Goncalves et al., 2022]. In parallel, uncertainty quantification methods such as Bayesian neural networks, MC dropout, and ensembles provide useful but imperfect epistemic uncertainty estimates [Blundell et al., 2015, Gal and Ghahramani, 2016, Lakshminarayanan et al., 2017, Lopez et al., 2025]. MC dropout is attractive because it is cheap, but it can be miscalibrated under dataset shift [Ovadia et al., 2019]. Our use of uncertainty is therefore not a standalone calibration method. The uncertainty estimate initializes evidential confidence, which becomes available for symbolic revision and case-level audit.

NARS and biomedical reasoning. NARS and Non-Axiomatic Logic model reasoning under insufficient knowledge and resources [Wang, 1995, 2006, 2022]. NAL truth values have been studied for uncertain medical diagnosis [Wang and Awan, 2011]. Biomedical knowledge graphs and symbolic reasoning can improve clinical interpretability [Wu et al., 2023, Abu-Salih et al., 2023, Rajabi and Kafaie, 2022, Jiang et al., 2024, Chudasama et al., 2025], but many such systems enrich representations rather than exposing an evidential confidence variable that directly controls neural inference. This paper uses NAL truth-value functions and rule-triggered revision only. It does not implement full NARS concept memory, task scheduling, resource allocation, or evidential-base management.

NGTA's contribution is its intervention boundary. Compared with post-hoc rule explanations, its rule path changes readout confidence and exports the numerical intervention. Compared with training-time knowledge injection, rules can be edited without retraining. Uncertainty calibration and ordinary gates remain essential controls; a shared truth-value interface does not establish new logic, independent evidence or clinical benefit.

3 Method

3.1 Interface Boundary

The implementation should be read as a NAL truth-value interface inspired by NARS, not as a full NARS system. It uses the (f, c) truth-value representation, deduction-style grounding of triggered rules, and revision of neural and symbolic judgments. It does not maintain concept memory, task queues, priority dynamics, resource-bounded control, or explicit evidential-base objects. The claims in this paper are therefore interface-level claims about truth-value-guided fusion and attention control.

3.2 Truth Values

For a statement S with positive evidence w^+ and negative evidence w^-, NAL represents truth as (f, c):

$$f = \frac{w^+}{w^+ + w^-}, \tag{1}$$

$$c = \frac{w^+ + w^-}{w^+ + w^- + k}. \tag{2}$$

We use the standard default $k = 1$, so $c = W/(W + 1)$ for total evidence $W = w^+ + w^-$. Strong deduction combines two premises by

$$f = f_1 f_2, \qquad c = f_1 c_1 f_2 c_2. \tag{3}$$

Revision merges two judgments with weights $\omega_i = c_i/(1 - c_i)$:

$$f = \frac{\omega_1 f_1 + \omega_2 f_2}{\omega_1 + \omega_2}, \qquad c = \frac{\omega_1 + \omega_2}{\omega_1 + \omega_2 + 1}. \tag{4}$$

Standard NAL revision assumes disjoint evidential bases [Wang, 2025]. The two software paths share clinical observations, so disjoint evidential bases are not established. Neural feature frequency is attention whereas symbolic frequency describes a clinical proposition; their semantic compatibility is unvalidated. Revised frequency is exported but does not affect the confidence-only gate.

3.3 Neural-to-Symbolic Initialization

For each feature, let p denote its mean attention weight and σ^2 its attention variance across MC-dropout passes. The initial feature truth value is

$$f_{\text{neural}} = p, \qquad c_{\text{neural}} = \frac{p(1-p) + \varepsilon}{p(1-p) + \sigma^2 + \varepsilon}. \tag{5}$$

Patient-level predictive probability and variance are mapped separately for reporting; they do not determine the feature gate. This is an application-specific initializer, not a canonical NARS measurement of evidential support. It parallels a variance-controlled evidence mass, then lets symbolic rules revise the confidence in NAL space. If the variance estimator underestimates uncertainty, c can be inflated. This limitation is inherited from the neural uncertainty method.

3.4 Symbolic Rule Revision

Each rule is a clinical proposition with a deterministic condition and a pre-specified symbolic truth value. A rule "fires" exactly when its condition evaluates true for the case. Firing does not depend on $(f_{\mathrm{sym}}, c_{\mathrm{sym}})$: f_{sym} encodes the prototype balance of supporting evidence and c_{sym} controls its evidential weight during revision. Triggered rules independently revise their matching feature truths by Eq. (4); any subset of rules may fire, and no conjunction of all four rules is required. The thresholds are interface probes, not learned decision thresholds or validated clinical cut points.

For an observed trigger, grounding uses observation frequency 1 and confidence 0.95. The implication confidence is reconstructed as $c_{\mathrm{sym}}/(0.95 f_{\mathrm{sym}})$ before strong deduction, so the deduction returns the preassigned symbolic truth. This records the chosen weight rather than independently estimating new evidence.

The symbolic truth values are prototype prior strengths, not an authoritative clinical knowledge base or literal outcome probabilities. A fired rule produces an inspectable event containing its identifier, condition, source feature, neural truth, symbolic truth, revised truth, and attention effect. A domain expert can therefore locate where a prior entered the model and can edit or disable that prior without retraining the transformer.

3.5 Attention Gating

Let α_i be the original attention weight for feature i and c_i the revised confidence associated with that feature. The gated attention is

$$\alpha_i^{\text{new}} = \frac{\alpha_i c_i^{\gamma}}{\sum_j \alpha_j c_j^{\gamma}}, \tag{6}$$

where γ controls the strength of confidence gating. The update preserves normalization; zero-confidence features receive zero gated weight unless all weights vanish, in which case the implementation retains the original attention. In this paper, $\gamma = 2$ is the default; a sensitivity run over $\gamma \in \{0.25, 0.5, 1, 2, 4\}$ is reported for TCGA.

3.6 Inference-Time Data Flow

The inference path has six stages. First, the trained transformer produces a point prediction, feature attention, and token scores. Second, repeated MC-dropout passes estimate predictive and attention variance. Third, the variance-to-confidence initializer in Eq. (5) creates feature-level neural truths. Fourth, the symbolic extractor emits every proposition whose condition is true. Fifth, neural and symbolic truths are revised in the shared (f, c) space. Sixth, revised confidence gates attention by Eq. (6); current implementation recomputes each cached pass's final logit and averages sigmoid probabilities identically across gate variants, then exports all events. The default intervention changes the token readout, not encoder representations. The CLS contribution remains ungated, and contextual tokens already mix features, so a zero readout weight does not remove an input's influence; a separate encoder intervention experiment recomputes those representations. There is no recurrent feedback or convergence loop.

This staged design keeps the symbolic mechanism outside model training. With N cases, R predicates, F features and T cached dropout passes, extraction and truth initialization cost $O(NR + NF)$; rescoring a readout gate costs $O(TNF)$, excluding transformer inference. The prototype assigns at most one rule to a logical feature and rejects collisions, so rule ordering cannot silently change a truth value. Larger bases need provenance-aware conflict policies, coverage analysis, and explicit aggregation of compatible or contradictory evidence before same-feature multi-rule revision can be supported.

3.7 Rule-Trace Semantics

For every triggered rule, the implementation records the feature name, rule condition, neural truth value, symbolic truth value, revised truth value, and resulting attention gain. These traces are not merely explanations after the fact. They correspond to actual arithmetic used by the prediction pipeline. A rule trace can therefore support three kinds of audit: whether the right rule fired, whether

its symbolic truth value is clinically defensible, and whether the revised confidence changed attention in the expected direction.

This is the main neurosymbolic distinction from a post-hoc explanation. In a post-hoc explanation, symbolic language may describe a neural decision without changing the computation. In dynamic evidential routing, symbolic truth values become part of the computation itself. The trace can still be wrong if the rule base is wrong or the proposition extractor is misaligned, but the failure is localizable.

3.8 Worked Trace

For an illustrative feature attention mean $p = 0.8$ and variance $\sigma^2 = 0.05$, and a fired prototype rule $(f_{\text{sym}}, c_{\text{sym}}) = (0.9, 0.7)$, Eq. (5) gives $(f_{\text{neural}}, c_{\text{neural}}) \approx (0.8000, 0.7619)$. Revision gives $(f_{\text{rev}}, c_{\text{rev}}) \approx (0.8422, 0.8469)$. In natural language, the rule contributes relatively strong positive evidence with moderate confidence; it does not mean a clinician assigned a 0.9 patient-level risk.

4 Experimental Setup

The current development evaluation fixes source hashes, partitions, rules, and training settings before each fit. WiDS contains 91,713 labeled stays with 65,737 training, 11,751 validation, and 14,225 held-out cases; patient, hospital, and ICU identifiers are disjoint across partitions. Five training seeds are evaluated under each combination of KNN or median imputation and APACHE included or excluded, yielding 20 fits. Removing APACHE leaves 15 encoded features instead of 16; score-only comparators remain available as controls.

TCGA contains 457 labeled cases with 319 training, 69 validation, and 69 held-out cases. An immutable GDC manifest pins 498 mutation files by publisher UUID, checksum, and case coverage; 16 expected clinical cases lack file-associated coverage. The fused input has 55 clinical and 100 genomic/unknown-indicator dimensions. Clinical-only, genomic-only, and fused inputs each receive five training seeds with identical partitions. Unknown mutation calls are not converted to negatives; no complete callable panel is verified. This is retrospective post-pathology association, and source-level measurement chronology is unresolved.

All fits use $d_{\text{model}} = 64$, four heads, two encoder layers, at most 60 epochs, and validation early stopping with patience 12. Each fit has 50 matched MC-dropout passes and three independent inference repeats. Calibration and classical-model selection use validation data only. These already inspected cohorts constitute development evaluation, not prospective confirmation or independent-cohort validation.

4.1 Baselines and Controls

The baseline transformer tests whether the underlying neural architecture is competitive before confidence gating is introduced. A uniform normalized readout gate cancels mathematically. Under matched current implementation aggregation it must reproduce ungated predictions within 10^{-7}. The legacy flat-control differences arise from aggregation, not from adding a gate. The MC-confidence-only ablation uses the neural variance-derived confidence but does not apply symbolic revision. The NARS-gated model adds the symbolic rule path on top of the MC-confidence initializer. This ordering is important: a difference from the baseline alone is not enough to show that symbolic reasoning helped, because confidence gating or regularization may explain the change.

The random forest baseline addresses the common tabular-learning concern that tree ensembles can be strong competitors on structured clinical features. It is trained on the same split and post-preprocessing design matrix, so differences cannot be attributed to a different data representation.

4.2 Symbolic Isolation, Uncertainty, and Shift

The matched evaluation rescores NAL revision after changing the symbolic input: removing every rule, removing one rule at a time, permuting rule assignments across patients, replacing symbolic truth values with random draws, and substituting fixed symbolic priors. Rule removal must reproduce MC-confidence-only gating. Sensitivity scales symbolic confidence and recomputes revision; it does not retune the transformer on test labels. A claim that the symbolic component improves prediction

Table 1: Prototype symbolic rules. Conditions are evaluated independently; any subset may fire. Truth values are evidential weights for the interface, not validated clinical probabilities. The frozen creatinine rule assumes mg/dL; the source dictionary lists micromol/L, an unresolved conflict.

Data	Rule	Trigger condition	(f, c)	Clinical reading
TCGA	BRAF mutation	BRAF mutation indicator = 1	$(0.85, 0.75)$	The rule increases support for the BRAF feature when the mutation is recorded; it is not a validated risk probability.
TCGA	Age >= 55	age at diagnosis >= 55 years	$(0.70, 0.60)$	The rule raises evidential support for the age feature without asserting that age alone determines metastasis.
TCGA	Pathologic T3/T4	pathologic T category is T3, T3a, T3b, T4, T4a, or T4b	$(0.90, 0.85)$	The rule increases support for the observed staging feature; it does not establish nodal spread by itself.
TCGA	Extrathyroid extension	extrathyroid extension is Minimal (T3), Moderate/Advanced (T4a), or Very Advanced (T4b)	$(0.85, 0.80)$	The rule raises support only for an explicitly recorded extension category; missing, placeholder, and unknown values do not fire.
WiDS	Lactate	maximum day-1 lactate >= 4.0 mmol/L	$(0.85, 0.80)$	The rule increases support for the lactate feature without diagnosing shock or determining mortality.
WiDS	Hypotension	minimum day-1 systolic blood pressure <= 90 mmHg	$(0.75, 0.70)$	The rule increases support for the blood-pressure feature; it is not a stand-alone mortality decision.
WiDS	Age	age >= 75 years	$(0.65, 0.60)$	The rule increases support for the age feature without treating age as sufficient for mortality.
WiDS	Creatinine	maximum day-1 creatinine >= 2.0 mg/dL	$(0.70, 0.65)$	The rule increases support for the creatinine feature without diagnosing kidney injury or determining mortality.

would require the stated rules to separate from MC-confidence-only gating and from these controls. That separation is not a result of the saved runs.

Each condition now has a five-model ensemble, MC-dropout variance, predictive entropy, confidence-error diagnostics, and tie-aware selective-risk curves. Common ensemble prediction errors provide the same reference for comparing uncertainty estimators. Hospital, training-seed, and dropout-repeat resampling distinguish these sources of variability; the fixed split does not estimate variability across cohort selections.

Each rule now carries a source note, a provenance identifier, an expert-review status, and a contradiction group. The current predicates are prototype probes and are marked not reviewed. Two rules that share a contradiction group and a target feature are rejected rather than overwritten. No independent cohort is evaluated here. The distribution-shift study masks originally observed raw values at 0, 10, 30, 50, and 70 percent before frozen preprocessing, rule extraction, and matched inference. Random, feature-dependent, and explicitly labeled outcome-dependent simulations share masks across methods. That check is a compatibility test under shift, not external clinical validation.

4.3 Rule Bases

The TCGA rule base contains four thyroid-cancer rules chosen to exercise distinct input channels: BRAF mutation for genomic evidence, age at least 55 years for demographic evidence, pathologic T3/T4 stage for staging evidence, and extrathyroid extension for pathology-derived evidence. The WiDS rule base contains four ICU rules chosen to exercise common tabular critical-care signals: lactate for metabolic stress, hypotension for hemodynamic instability, age for demographic risk, and creatinine for renal dysfunction. All rules are applied only at inference time.

The small rule bases are prototype probes, not complete clinical ontologies. Clinical use would require expert-curated propositions, population-specific threshold rationales, provenance, and sensitivity analysis. Existing guideline or score components could be encoded and run in shadow mode to measure coverage, disagreement with extracted data, and downstream model effects. Such experiments could reveal where a rule interacts with the model; they would not validate the clinical correctness of the rule itself. Symbolic revision would be empirically isolated only if it separated from baseline, flat-confidence, and MC-confidence controls in paired comparisons. The current results do not meet that standard.

4.4 Statistical Reporting

Current performance tables report five-seed means and seed standard deviations; the performance figure uses Student-t intervals over seed means. Paired symbolic contrasts resample held-out hospitals or TCGA cases, training seeds and dropout repeats. Secondary comparisons retain their family adjustments. Observed deltas are reported separately from bootstrap centers. These fixed-split development intervals do not establish prospective confirmation, clinical equivalence or cohort-selection uncertainty.

Table 2: Current development evaluation: five training seeds per condition on fixed held-out cases. Brier values are means with seed standard deviations, not case-bootstrap confidence intervals. The APACHE score remains a score-only comparator when removed from predictor inputs.

Condition	Variant	AUROC	Brier (seed SD)
WiDS KNN	Ungated	0.875721	0.057830 (0.000109)
WiDS KNN	MC-only	0.875711	0.057828 (0.000112)
WiDS KNN	NARS	0.875713	0.057828 (0.000112)
WiDS KNN	Boosting	0.873278	0.057864 (0.000156)
WiDS KNN	Recalibrated	0.875721	0.057782 (0.000145)
TCGA fused	Ungated	0.692941	0.229410 (0.009980)
TCGA fused	MC-only	0.692101	0.229534 (0.010085)
TCGA fused	NARS	0.692101	0.229548 (0.010068)
TCGA fused	Boosting	0.661345	0.243924 (0.000000)
TCGA fused	Recalibrated	0.692941	0.226140 (0.008783)

Table 3: Matched symbolic contrasts, NARS-minus-MC Brier, with hierarchical 95% intervals over cases/hospitals, training seeds, and three dropout repeats. Negative is better. These development comparisons do not meet the prespecified 10^{-4} minimum benefit.

Condition	Difference	Lower	Upper
WiDS KNN	-2.43e-07	-6.91e-07	1.64e-07
WiDS median	-4.04e-07	-1.19e-06	5.79e-07
WiDS KNN, no APACHE	-1.68e-07	-4.31e-07	3.11e-07
WiDS median, no APACHE	-2.35e-07	-7.85e-07	6.53e-07
TCGA fused	1.26e-05	-5.69e-05	1.06e-04
TCGA clinical	2.63e-05	-5.74e-05	1.33e-04
TCGA genomic	-1.04e-05	-4.35e-05	8.06e-06

The paper therefore separates three evidential claims. First, whether the architecture compiles and runs on the benchmark. Second, whether the symbolic pathway fires and changes feature-level truth values. Third, whether the symbolic pathway changes predictive metrics beyond simpler controls. The present results support the first two claims. They do not isolate the third claim. This separation is central to the paper's framing.

4.5 Reproducibility Interface

The repository supplies a frozen 36-fit recipe, a dry-run/execute reproduction command, integrity verification and a source-only package builder. The package excludes source datasets, fitted arrays, patient-level traces and private study materials. Exact replay needs the locally retained fitted artifacts and authorized source data; the source package alone supports regeneration rather than access to restricted records. The paper text reports the split sizes, input dimensions, architecture, MC-dropout count, and baselines needed to interpret the results. The code is still necessary for exact reproduction because preprocessing, imputation, categorical encoding, and rule extraction are implementation-sensitive.

In current implementation, imputation, standardization, categorical encoding, sparse-column filtering, and mutation-panel selection fit training data only. Legacy sparse filtering and panel selection occurred before splitting. Unknown gene calls now have explicit missing indicators; confirmed negative calls require sourced case/gene assay coverage. TCGA selects one physical record per source table and exports conflicts, but cross-table chronology is unverified; its landmark is retrospective post-pathology association, not preoperative prediction. The rule extractor is also run only at inference time, so symbolic rules do not alter the supervised training objective.

5 Current Development Results

The seven conditions in Table 3 do not establish the planned symbolic benefit. WiDS KNN NARS-minus-MC Brier is -2.43×10^{-7} with hierarchical 95% interval $[-6.91 \times 10^{-7}, 1.64 \times 10^{-7}]$. TCGA fused routing has a positive mean difference of 1.26×10^{-5}, with an interval spanning zero. Direct confidence boosts, frequency sensitivity, individual/all-rule removal, 100 predicate permutations, shuffled confidence assignments, and matched irrelevant predicates are retained as symbolic controls.

Table 4: Operational auditability for each training seed on the same held-out cohort. Events include unmapped triggers; finite values and confidence changes refer to mapped events. Rows repeat cases across fits and must not be added. Residuals measure numerical fidelity, not clinician usability or rule validity.

Data	Cases covered	All (mapped)	Finite	c changed	Flips	Max residual
TCGA-THCA seed_0	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
TCGA-THCA seed_1	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
TCGA-THCA seed_2	56/69 (81.2%)	113 (89)	100.0%	100.0%	2	4.4e-16
TCGA-THCA seed_3	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
TCGA-THCA seed_4	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
WiDS ICU seed_0	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	3	3.3e-16
WiDS ICU seed_1	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	2	4.4e-16
WiDS ICU seed_2	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	3	4.4e-16
WiDS ICU seed_3	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	0	4.4e-16
WiDS ICU seed_4	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	1	4.4e-16

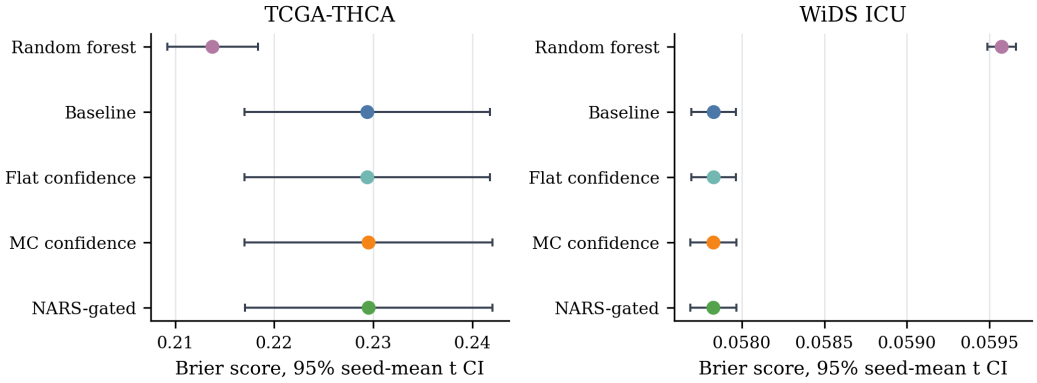


Figure 2: Current five-seed mean Brier scores with 95% Student-t intervals over training seeds on fixed held-out cases. These are seed-mean intervals, not case-bootstrap intervals or external confirmation.

Revised frequency does not determine the confidence gate, and shared input measurements do not establish independent evidence.

Mean WiDS ungated AUROC is 0.875721 with KNN and APACHE, versus 0.841118 without APACHE; median yields 0.874062 versus 0.842176. Paired hospital-and-seed contrasts preserve common test cases and report multiplicity-adjusted intervals. Boosting beats the median transformer in mean Brier, and validation recalibration can exceed the small routing effects. These results expose dependence on an existing mortality score rather than establish added clinical value over the strongest comparator. Authoritative APACHE sentinel semantics and timestamp availability remain unverified.

A newly obtained WiDS dictionary labels creatinine in micromol/L, whereas the frozen prototype rule assumes mg/dL. We therefore report a rule-off sensitivity without choosing a replacement unit or threshold. Across all 20 WiDS fits, disabling that revision reproduces the saved individual-rule control within 10^{-12} Brier and changes no 0.5-threshold decisions. All four hospital-and-seed intervals for rule-off-minus-MC Brier span zero. This preserves the negative symbolic conclusion but does not resolve the source-unit conflict; clinical interpretation and external harmonization remain deferred.

The acquired TCGA held-out panel has recorded positives in 50 of 69 cases and produces 39 BRAF, 29 T-stage, 24 extension, and 21 age events per seed. All 113 events are exported; 89 map to fitted features and 24 are explicitly unmapped. WiDS observed-only rules produce 11,457 mapped events in 8,152 of 14,225 cases per seed. Numerical replay validates predicates, truth revision, gates, probabilities, and rule-off effects; it does not certify the clinical predicates. Seed rows reuse the same cases and must not be combined as independent patient counts.

Controlled missingness outputs preserve frozen preprocessing, rules, model weights, and matched dropout inference. Per-fit paired intervals test integrated degradation and the 0.001 per-mask harm margin; uncertainty diagnostics include error-detection AUROC and selective-risk area with hospital-

clustered intervals. The criterion passes in 12 of 15 median seed/scenario combinations and 1 of 15 KNN combinations; no KNN random or feature-dependent scenario passes. This does not establish general routing robustness. Outcome-dependent masks are simulations, not observed missingness mechanisms, and numerical acceptance under a particular simulation does not establish population robustness.

For the fixed five-model WiDS KNN ensemble, mean predictive MC variance detects common threshold errors with AUROC 0.842276 and hospital-bootstrap interval [0.833229, 0.857306]. Neural and revised attention-instability scores give 0.514950 [0.493533, 0.531916] and 0.500647 [0.477047, 0.515520], respectively. Their intervals span chance: informative predictive uncertainty does not establish the attention confidence as a useful trust signal. These conditional intervals exclude model-refitting variability.

The reviewer protocol, power sensitivity and crossed reviewer/case analysis are implemented. A fresh private package now randomizes fault assignment, uses opaque case IDs and isolates each reviewer's assigned format; it contains 1,920 balanced assignments for 24 planned reviewers and 80 evaluation cases. No participants were recruited, no responses were generated as human evidence, and no reviewer benefit is claimed. External evaluation similarly requires a documented independent cohort with matched units, windows, outcomes, and patient identities; no such cohort was supplied.

Historical corrections. Earlier unmatched-aggregation tables and sparse-source genomic demonstrations are retained in the repository archive, not used as current evidence. Their apparent calibration differences did not isolate a gate effect; the earlier genomic test split had no observed mutations, imputed values could trigger rules, and the revision audit reused production arithmetic. The present matched evaluation and independent persisted replay address these implementation errors without establishing clinical benefit.

6 Discussion and Ablation

The flat-confidence and MC-confidence-only controls are essential for interpreting the result. If NARS-gated attention were compared only with the baseline transformer, its paired Brier and ECE differences could be over-interpreted as a symbolic reasoning effect. Uniform identity removes the aggregation confound in the current experiments, while MC-only and matched arbitrary-rule controls isolate the additional rule-conditioned confidence change. The NARS rule path is therefore best interpreted as an instrumentation layer with inspectable numerical activity and editable prototype rules, not yet a statistically isolated source of within-family calibration change.

Figure 3 uses the current five-seed TCGA fused and WiDS KNN sources. Gamma sweeps and decision curves are descriptive development diagnostics; no gamma is selected using test outcomes, and overlapping net-benefit curves do not establish clinical utility.

6.1 Failure Modes and Deployment Boundary

The interface can fail when MC dropout underestimates uncertainty, extraction misaligns features with clinical concepts, or rule truth values are disputed. Traces expose extracted conditions and revisions for inspection and rule-off replay, but whether this instrumentation helps human oversight has not been evaluated.

The current system remains a methods prototype rather than a clinical decision device. Deployment would require a rule base curated independently by domain experts, external validation without retuning, shift monitoring, rule governance, human-factors evaluation, and clinician override. Stronger evidence for symbolic revision would also require separation from baseline, flat-confidence, and uncertainty-only controls across external cohorts.

7 Limitations

The inspected development cohorts contain 69 held-out TCGA cases and WiDS hospitals within one benchmark. Both rule bases lack expert review. Ensemble and masking intervals omit cohort-selection uncertainty; MC dropout may understate uncertainty under shift. Clinical validation remains unestablished.

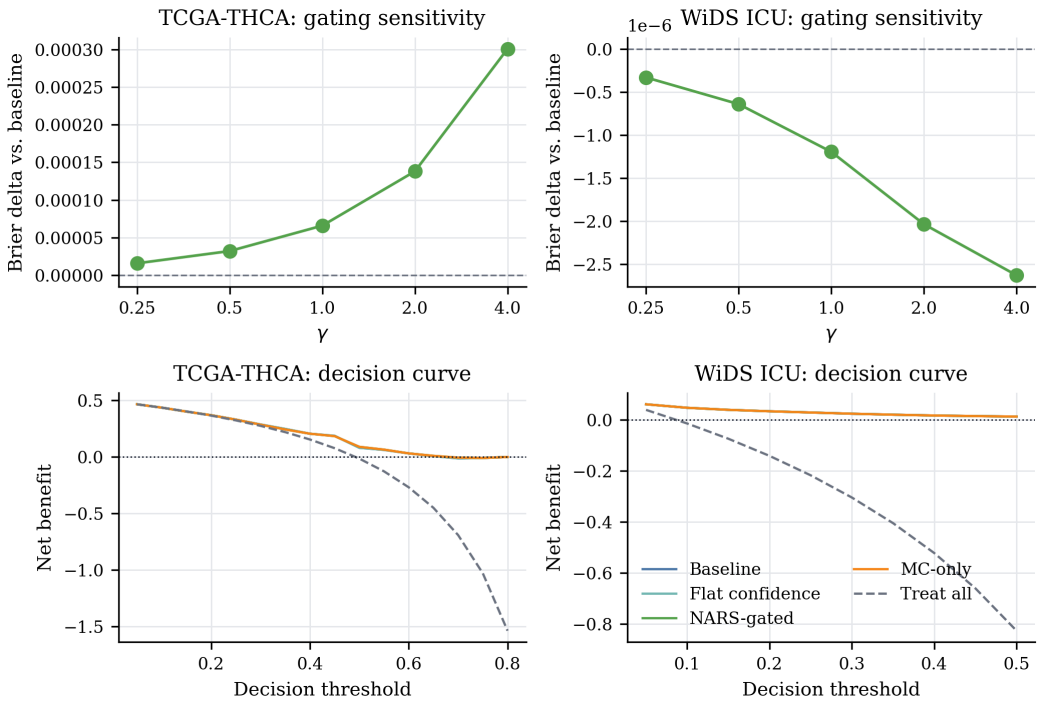


Figure 3: Reported ablations. Top: NARS-gated Brier change relative to the ungated transformer across γ (negative is lower). Bottom: held-out decision curves over informative threshold ranges. These descriptive curves do not establish clinical utility.

An exploratory sensitivity evaluates ten frozen no-APACHE checkpoints on 12,000 PhysioNet/CinC 2012 records[1], with first-day inputs and complete replay. Missing SpO2/surgery inputs, original $\geq$ 48-hour stay selection, and unverified identity/window equivalence prevent independent clinical claims; intervals condition on one institution. TCGA pathology dates remain absent in 507 selected records. All 1,554 pinned public GDC XML supplements across 507 cases were also audited, alongside the biotabs. Specimen dates include 515 procurement/review offset pairs, but do not verify selected clinical/genomic input availability or gene callability.

NGTA implements NAL truth-value functions and feature-level revision, without NARS concept priorities, task scheduling, or explicit evidential-base records. Its claims concern an inference interface, not a complete NARS architecture.

8 Conclusion

Dynamic evidential routing is an auditable inference-time interface for uncertainty-conditioned symbolic intervention. Explicit rules can revise confidence and change attention, and the run leaves arithmetic traces. The saved comparisons do not show that symbolic revision improves clinical prediction. External cohorts, expert-curated rules, and human evaluation are needed before clinical use.

Ethics and reproducibility. This study uses existing de-identified benchmarks and collected no new patient data. The WiDS publisher requires registered access and a data-use agreement. The reproduction package includes code, frozen configurations, tests and aggregate paper assets; authorized data and private study materials remain separate. Large language models assisted with drafting, formulation, and code generation; the authors checked the derivations, implementation, references, and results and take responsibility for the work.

[1]Source: `https://physionet.org/content/challenge-2012/1.0.0/`.

References

Bilal Abu-Salih, Muhammad AL-Qurishi, Mohammed Alweshah, Mohammad AL-Smadi, Reem Alfayez, and Heba Saadeh. Healthcare knowledge graph construction: A systematic review of the state-of-the-art, open issues, and opportunities. *Journal of Big Data*, 10(1):83, 2023. doi: 10.1186/s40537-023-00774-9.

Shekoofeh Azizi, Basil Mustafa, Fiona Ryan, Zachary Beaver, Jan Freyberg, Jonathan Deaton, Aaron Loh, Alan Karthikesalingam, Simon Kornblith, Ting Chen, Vivek Natarajan, and Mohammad Norouzi. Big self-supervised models advance medical image classification. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 3478–3488, 2021. doi: 10.1109/iccv48922.2021.00346. URL https://openaccess.thecvf.com/content/ICCV2021/html/Azizi_Big_Self-Supervised_Models_Advance_Medical_Image_Classification_ICCV_2021_paper.html.

Bikram Pratim Bhuyan, Amar Ramdane-Cherif, Ravi Tomar, and T. P. Singh. Neuro-symbolic artificial intelligence: A survey. *Neural Computing and Applications*, 36(21):12809–12844, 2024. doi: 10.1007/s00521-024-09960-z.

Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight uncertainty in neural network. In *Proceedings of the International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1613–1622, 2015. URL https://proceedings.mlr.press/v37/blundell15.html.

Yashrajsinh Chudasama, Hao Huang, Disha Purohit, and Maria-Esther Vidal. Towards interpretable hybrid AI: Integrating knowledge graphs and symbolic reasoning in medicine. *IEEE Access*, 13: 39489–39509, 2025. doi: 10.1109/access.2025.3529133.

Brandon C. Colelough and William Regli. Neuro-symbolic AI in 2024: A systematic review. arXiv:2501.05435, 2025. URL https://arxiv.org/abs/2501.05435.

Lixia Dai, Hang Xu, and Yugui Zhang. Automated classification of clinical diagnoses in electronic health records using transformer. *PLOS One*, 20(9):e0329963, 2025. doi: 10.1371/journal.pone.0329963.

Artur d'Avila Garcez and Luís C. Lamb. Neurosymbolic AI: The 3rd wave. *Artificial Intelligence Review*, 56(11):12387–12406, 2023. doi: 10.1007/s10462-023-10448-w.

Artur d'Avila Garcez, Marco Gori, Luis C. Lamb, Luciano Serafini, Michael Spranger, and Son N. Tran. Neural-symbolic computing: An effective methodology for principled integration of machine learning and reasoning. *Journal of Applied Logics – IfCoLog Journal of Logics and their Applications*, 6(4):611–631, 2019. URL https://openaccess.city.ac.uk/id/eprint/22984/.

Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–1059, 2016. URL https://proceedings.mlr.press/v48/gal16.html.

Marzyeh Ghassemi, Tristan Naumann, Peter Schulam, Andrew L. Beam, Irene Y. Chen, and Rajesh Ranganath. A review of challenges and opportunities in machine learning for health. *AMIA Joint Summits on Translational Science Proceedings*, 2020:191–200, 2020. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC7233077/.

Tiago Goncalves, Isabel Rio-Torto, Luis F. Teixeira, and Jaime S. Cardoso. A survey on attention mechanisms for medical applications: Are we moving toward better algorithms? *IEEE Access*, 10: 98909–98935, 2022. doi: 10.1109/access.2022.3206449.

M. Hilario. An overview of strategies for neurosymbolic integration. In Ron Sun and Frederic Alexandre, editors, *Connectionist-Symbolic Integration: From Unified to Hybrid Approaches*. Lawrence Erlbaum Associates, 1997. URL https://www.routledge.com/Connectionist-Symbolic-Integration-From-Unified-to-Hybrid-Approaches/Sun-Alexandre/p/book/9780805823493.

Pengcheng Jiang, Cao Xiao, Minhao Jiang, Parminder Bhatia, Taha Kass-Hout, Jimeng Sun, and Jiawei Han. Reasoning-enhanced healthcare predictions with knowledge graph community retrieval. arXiv:2410.04585, 2024. URL https://arxiv.org/abs/2410.04585.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html.

Fei Li, Yonghao Jin, Weisong Liu, Bhanu Pratap Singh Rawat, Pengshan Cai, and Hong Yu. Fine-tuning bidirectional encoder representations from transformers (BERT)-based models on large-scale electronic health record notes: An empirical study. *JMIR Medical Informatics*, 7(3):e14830, 2019a. doi: 10.2196/14830.

Xuedong Li, Yue Wang, Dongwu Wang, Walter Yuan, Dezhong Peng, and Qiaozhu Mei. Improving rare disease classification using imperfect knowledge graph. *BMC Medical Informatics and Decision Making*, 19(S5):238, 2019b. doi: 10.1186/s12911-019-0938-1.

Leopoldo Julian Lechuga Lopez, Shaza Elsharief, Dhiyaa Al Jorf, Firas Darwish, Congbo Ma, and Farah E. Shamout. Uncertainty quantification for machine learning in healthcare: A survey. In *Proceedings of the sixth Conference on Health, Inference, and Learning*, volume 287 of *Proceedings of Machine Learning Research*, pages 862–907. PMLR, 2025. URL https://proceedings.mlr.press/v287/lopez25a.html.

Uzma Nawaz, Mufti Anees ur Rahaman, and Zubair Saeed. A review of neuro-symbolic AI integrating reasoning and learning for advanced cognitive systems. *Intelligent Systems with Applications*, 26:200541, 2025. doi: 10.1016/j.iswa.2025.200541.

Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html.

Enayat Rajabi and Somayeh Kafaie. Knowledge graphs and explainable AI in healthcare. *Information*, 13(10):459, 2022. doi: 10.3390/info13100459.

Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. Med-BERT: Pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine*, 4(1):86, 2021. doi: 10.1038/s41746-021-00455-y.

Pei Wang. *Non-Axiomatic Reasoning System: Exploring the Essence of Intelligence*. PhD thesis, Indiana University, 1995. URL https://cis.temple.edu/~wangp/papers.html.

Pei Wang. *Rigid Flexibility: The Logic of Intelligence*. Springer, 2006. URL https://cis.temple.edu/~wangp/papers.html.

Pei Wang. Intelligence: From definition to design. In *Proceedings of the Third International Workshop on Self-Supervised Learning*, volume 192 of *Proceedings of Machine Learning Research*, pages 35–47. PMLR, 2022. URL https://proceedings.mlr.press/v192/wang22a.html.

Pei Wang. *Non-Axiomatic Logic: A Model of Intelligent Reasoning*. World Scientific, 2 edition, 2025. doi: 10.1142/14486. URL https://cis.temple.edu/~wangp/papers.html.

Pei Wang and Seemal Awan. Reasoning in non-axiomatic logic: A case study in medical diagnosis. In *Proceedings of the Conference on Artificial General Intelligence*, pages 297–302, 2011. doi: 10.1007/978-3-642-22887-2_33.

Xuehong Wu, Junwen Duan, Yi Pan, and Min Li. Medical knowledge graph: Data sources, construction, reasoning, and applications. *Big Data Mining and Analytics*, 6(2):201–217, 2023. doi: 10.26599/bdma.2022.9020021.